

Training Day – 16

Today's training focused on Jenkins and its role in Continuous Integration and Continuous Delivery/Deployment (CI/CD). The session provided practical experience with Jenkins projects and pipelines while introducing Groovy scripting and Jenkinsfiles.

Work Undertaken:

- Learned what **Jenkins** is and understood how it automates various stages of CI/CD workflows.
- Understood the relationship between **GitHub and Jenkins**: GitHub is used to store and manage source code, while Jenkins can monitor repositories and automate tasks such as building, testing, and deploying applications.
- Studied the importance of Jenkins in automating repetitive **software development and deployment tasks**, reducing manual effort and improving consistency.
- Created **Jenkins Freestyle Projects** and practiced configuring basic automated jobs.
- Implemented basic **Jenkins Pipelines** to understand how multiple stages of a CI/CD workflow can be defined and executed.
- Learned the fundamentals of **Groovy**, the scripting language commonly used for defining Jenkins pipeline logic.
- Understood the purpose of a **Jenkinsfile**, which defines the pipeline configuration and instructions that Jenkins should execute.
- Learned that a **Jenkinsfile** is typically stored in the project's root directory so that Jenkins can locate and execute the pipeline configuration directly from the project repository.

Additional Learning:

The session helped connect version control with automated software delivery. Understanding how Jenkins interacts with GitHub demonstrated how source-code changes can trigger automated workflows for building, testing, and deploying applications.

Conclusion:

The day provided valuable practical knowledge of **Jenkins and CI/CD automation**. Creating Freestyle Projects and basic Pipelines, along with learning Groovy and Jenkinsfiles, established a foundation for implementing automated software development and deployment workflows.